Journal of Business and Psychology, Vol. 18, No. 4, Summer 2004 (© 2004)

UNCERTAINTY DURING ORGANIZATIONAL CHANGE: TYPES, CONSEQUENCES, AND MANAGEMENT STRATEGIES

Prashant Bordia

Elizabeth Hobman

University of Queensland

Elizabeth Jones

Griffith University

Cindy Gallois

Victor J. Callan

University of Queensland

ABSTRACT: This research tested a model that classifies change uncertainty into three interrelated types: strategic, structural, and job-related. We predicted that control would mediate the effects of job-related uncertainty upon psychological strain, and that management communication and participation in decision-making (PDM) would reduce uncertainty and increase feelings of control. The model was tested in a public sector organization and the results supported it. Control was found to mediate the effects of job-related uncertainty upon psychological strain. Management communication was negatively related to strategic uncertainty, whereas PDM was negatively related to structural and job-related uncertainty, suggesting different mechanisms to deal with the types of uncertainty during change. Finally, PDM was positively associated with feelings of control and negatively associated with psychological strain. These results suggest that PDM can short-circuit the damaging effects of uncertainty by allowing employees

---

This research was supported by a grant from the Australian Research Council. An earlier version of this paper was presented at the 2002 International Congress of Applied Psychology, Singapore. We thank Nicholas DiFonzo and Nerina Jimmieson for their comments on this manuscript.

Address correspondence to Prashant Bordia, School of Psychology, University of Queensland, St. Lucia, QLD 4072, Australia. E-mail: prashant@psy.uq.edu.au.

0889-3268/04/0600-0507/0

to have a say in change related organizational affairs, thereby instilling a sense of control over their circumstances.

KEY WORDS: uncertainty; organizational change; communication.

In today's continually changing business environment, often organizations have to change strategic direction, structure and staffing levels to stay competitive (Armenakis & Bedeian, 1999; Cascio, 1995). These changes lead to a great deal of uncertainty and stress among employees (Callan, 1993; Terry & Jimmieson, 2003). There is a growing literature on the nature and consequences of uncertainty during organizational change (DiFonzo & Bordia, 1998; Maurier & Northcott, 2000; Nelson, Cooper, & Jackson, 1995; Pollard, 2001; Rafferty, 2002; Schweiger & Denisi, 1991; Terry, Callan, & Sartori, 1996). While this literature has noted the presence of uncertainty and its negative consequences for psychological well-being, there is limited research on the precise psychological mechanism explaining the negative consequences of uncertainty. In this study, we developed and tested a model describing the different types of uncertainties experienced during organizational change, their consequences for psychological well-being, and strategies to manage uncertainty. The literature review is divided into three broad sections. First, the literature on uncertainty during organizational change is reviewed. We integrate the different types of uncertainties and propose a three-factor conceptualization of uncertainty. Second, we review the literature on consequences of uncertainty and propose that the relationship between uncertainty and strain is mediated by feelings of control. Finally, we present communication and participation in decision making as strategies aimed at managing uncertainty and its negative consequences.

UNCERTAINTY DURING ORGANIZATIONAL CHANGE

Uncertainty has been defined as "an individual's inability to predict something accurately" (Milliken, 1987, p. 136). This could be due to lack of information (Berger & Calabrese, 1975) or ambiguous and contradictory information (Putnam & Sorenson, 1982). However, a characteristic feature of uncertainty is the sense of doubt about future events or about cause and effect relationships in the environment (DiFonzo & Bordia, 1998). Uncertainty is an aversive state that motivates strategies aimed at reducing or managing it. This idea has been a central tenet in several theoretical approaches in psychology and communication. For example, uncertainty reduction is the primary motivation for group identification

(Hogg & Mullin, 1999), social comparison (Festinger, 1954), rumor activity (DiFonzo, Bordia, & Rosnow, 1994; Rosnow, 1991) and information seeking in interpersonal (Berger & Bradac, 1982) and organizational contexts (Ashford & Black, 1996; Kramer, 1999; Morrison, 2002).

Uncertainty is one of the most commonly reported psychological states in the context of organizational change. For example, during a merger employees may experience uncertainty about the nature and form of the merged organization, impact of the merger on their work unit and the likely changes to their job role (Bastien, 1987; Buono & Bowditch, 1989; DiFonzo & Bordia, 1998; Terry et al., 1996). Similarly, in times of organizational restructuring, employees feel uncertain about the changing priorities of the organization and the likelihood of lay-offs. Thus, there are a large number of issues about which employees may feel uncertain. One measure of uncertainty during organizational change (Schweiger & Denisi, 1991) listed 21 sources of uncertainty, including uncertainty regarding lay-offs, pay cut, promotion opportunities, and changes to the culture of the organization. The wide variety of uncertainties poses a challenge to managers developing communication and change implementation strategies aimed at minimizing uncertainty.

There have been some attempts at developing taxonomies of uncertainties in the organizational context. Jackson, Schuler, and Vredenburgh (1987) classified uncertainty at three levels of analysis: organizational (e.g., uncertainty about external business environments), group (e.g., uncertainty regarding structure of the organization), and individual (e.g., job, task, and role related uncertainty). Similarly, Buono and Bowditch (1989) argued that sources of uncertainty could be classified into three levels: external (including environmental uncertainty due to technological and market changes), organizational (uncertainty due to changes to the organizational structure and culture) and individual (uncertainty regarding job role and status).

Adapting the approaches used by Buono and Bowditch (1989) and Jackson et al. (1987), we proposed a three-factor conceptualization of uncertainty during change comprising strategic, structural, and job-related uncertainty. While our taxonomy is a close adaptation of that proposed by Jackson et al. (1987), there is one important difference. Jackson et al.'s taxonomy operated at multiple levels of analysis (organizational, group, and individual). Our taxonomy, however, is applied only to the individual level of analysis. Thus, we are interested in the subjective experience or appraisal of different uncertainties by *individuals* in a changing organization, and we suggest that the different types of uncertainties can be grouped into three factors: strategic, structural, and job-related uncertainties.

We describe the three types of uncertainties below, along with illustrative extracts. The extracts were taken from interviews conducted as part of a research project on employee reactions to organizational change in a healthcare facility undergoing large-scale redevelopment (partial privatization and relocation to a new building). In addition to uncertainty, a variety of attitudinal and affective reactions were identified. Detailed results of these interviews are reported elsewhere (Watson, Jones, Hobman, Bordia, Gallois, & Callan, 2002). However, we have used quotes related to uncertainty to illustrate our conceptualization of the strategic, structural, and job-related uncertainties.

In our model, strategic uncertainty refers to uncertainty regarding organization-level issues, such as reasons for change, planning and future direction of the organization, its sustainability, the nature of the business environment the organization will face, and so forth. Sias and Wyers (2001), in a study on uncertainty among employees of a newly formed business, found that uncertainty regarding the future viability of the organization was particularly high. Similarly, changes in government create uncertainty among public service personnel regarding the impact of changing policies (e.g., privatization, funding cuts) on an organization's strategic direction (Desveaux, 1994). In the context of change, staff may feel uncertain regarding the reasons for change or the overall nature of change. For example, in the words of an employee from the healthcare facility described above: "We know we are being redeveloped. We don't know which direction we're going in terms of the redevelopment." The uncertainty often reflects a lack of clear vision or strategic direction by the leaders of change (Kotter, 1996).

There is a substantial literature on environmental uncertainty and strategic decision making, but this literature primarily focuses on top management's response to dynamic or unpredictable business environments (Milliken, 1990). Our conceptualization of strategic uncertainty specifically concerns the context of changing organizations and the experience of uncertainty among all staff, not just top management. The term strategic, rather than environmental uncertainty, is preferred because this definition is broader and includes uncertainty about reasons for change and the future viability of the organization, in addition to uncertain business environments.

The second element of our conceptualization, structural uncertainty, refers to uncertainty arising from changes to the inner workings of the organization, such as reporting structures and functions of different work-units. Organizational restructuring often involves merging of work units, disbanding of unprofitable departments, and team-based restructuring. These changes create uncertainty regarding the chain of command, relative contribution and status of work units, and policies and practices (Buono & Bowditch, 1989). With mergers, for instance, there is often a combination of business units, reallocation of services and staff are physically relocated to different parts of the organization (Terry et al., 1996). One hospital employee expressed uncertainty about the impact of redevelopment upon the structure and nature of a work unit: "We still don't know whether they'll refurbish the hydrotherapy pool as part of the redevelopment. Um . . . there's a lot of uncertainty over what will happen with spinal injuries unit." Similarly, a nurse expressed uncertainty regarding relocation of work units and changing reporting structures: "the nursing staff at Unit X . . . [are] attached to [Hospital A] at the moment. That's who . . . they're accountable to. And yet they're, they're moving Unit X to [Hospital B]. . . . So the nursing staff don't know whether they're going to be accountable to [Hospital A]. They're currently on site at [Hospital C]. They don't know whether they're going to be accountable to [Hospital C] or . . . whether they're going to be accountable to [Hospital B]." Structural uncertainty can operate at both the vertical and horizontal levels of the organization. While change in the 1990s was mostly about de-layering and downsizing middle layers of management and associated roles, there is now a greater focus upon horizontal restructuring. Here the intention is to break down silos between business units and to create value-adding to the services and products being produced (Carnall, 1999).

Finally, job-related uncertainty includes uncertainty regarding job security, promotion opportunities, changes to the job role, and so forth. Job-related uncertainties are widely prevalent in changing organizations and have been extensively noted in the literature (Bastien, 1987; DiFonzo & Bordia, 1998; Ito & Brotheridge, 2001; Maurier & Northcott, 2000; Nelson et al., 1995; Schweiger & Denisi, 1991). Indeed, of the 21 different sources of uncertainty in Schweiger and Denisi's measure, 18 were specifically about job-related issues. Changes in structure or design of organizations, introduction of new technology, and downsizing programs lead to changes to job roles and create job-related uncertainty and insecurity (Cascio, 1995; Ito & Brotheridge, 2001). In the words of a hospital employee: "We don't know if we're going. I mean they can't tell us that we're going to be definitely [losing our job], all they will tell us actually is [that] the job [we are] doing now we will not be doing in the new hospital."

The classification of change-related uncertainties into the three broad types helps us understand how the different types of uncertainties during change might be related. Most processes in organizations are highly interdependent, and the strategic, structural, and job-related aspects of an organization are often nested sub-systems (Jackson et al., 1987). The three types of uncertainties can affect each other. The direction of influence is likely to be from the higher levels to the lower level, in a cascade-like fashion (Jackson et al., 1987). That is, strategic uncertainty is likely to lead to structural uncertainty which, in turn, contributes to job-related uncertainty. For example, media reports of talks between CEOs of two organizations might lead to merger speculation and create strategic uncertainty. This would lead to uncertainty about the shape and form of the merged organization or relative status of two competing work units in the merging organizations (structural uncertainty). This would, in turn, create uncertainty about changes to job roles, the need for re-training, job transfers, or even lay-offs (job-related uncertainty). Therefore, we made the following predictions about the relationship between the three types of uncertainties:

Hypothesis 1: Strategic uncertainty is positively related to structural uncertainty.

Hypothesis 2: Structural uncertainty is positively related to job-related uncertainty.

CONSEQUENCES OF UNCERTAINTY

Uncertainty has several negative consequences for individual wellbeing and satisfaction in the organizational context. It is positively associated with stress (Ashford, 1988; Pollard, 2001; Schweiger & Denisi, 1991) and turnover intentions (Greenhalgh & Sutton, 1991; Johnson, Bernhagen, Miller, & Allen, 1996) and negatively associated with job satisfaction (Ashford, Lee, & Bobko, 1989; Nelson, Cooper, & Jackson, 1995), commitment (Ashford et al., 1989; Hui & Lee, 2000), and trust in the organization (Schweiger & Denisi, 1991).

The negative consequences of uncertainty for psychological wellbeing are largely due to the feelings of lack of control that uncertainty engenders (Bordia, Hunt, Paulsen, Tourish, & DiFonzo, 2001; DiFonzo & Bordia, 2002; Lazarus & Folkman, 1984). Control has been defined as "an individual's beliefs, at a given point in time, in his or her ability to effect a change, in a desired direction, on the environment" (Greenberger & Strasser, 1986, p. 165). Uncertainty, or lack of knowledge about current or future events, undermines our ability to influence or control these events. This lack of control, in turn, leads to negative consequences, such as anxiety (DiFonzo & Bordia, 2002), psychological strain (Spector, 1986, 1987, 2002; Terry & Jimmieson, 1999), learned helplessness (Martinko & Gardner, 1982), and lower performance (Bazerman, 1982; Jimmieson & Terry, 1999; Orpen, 1994). The mediating role of control in the uncertainty—strain relationship has been theorized in the psychology, communication, and organizational literatures. According to the stress appraisal and coping model (Lazarus & Folkman, 1984), a primary appraisal of uncertainty is followed by the secondary appraisal of personal control before coping resources are activated. Similarly, information-seeking strategies by new employees are aimed at reducing uncertainty and thereby asserting control over their work environment (Ashford & Black, 1996). Control can mediate the relationship between uncertainty and anxiety (DiFonzo & Bordia, 2002) and between uncertainty and psychological strain (Bordia et al., 2001). Therefore, we predicted that uncertainty would be negatively related to control, which in turn, would be negatively related to psychological strain.

It is likely that different types of uncertainties have different effects upon employee well-being (Jackson et al., 1987). Of the three different types of uncertainties described above, we expected job-related uncertainties to be the most stressful, as job-related issues are of greatest personal relevance to employees (Klein, 1996). In line with the cascade effect outlined above, we expected that the effects of strategic and structural uncertainty on lack of control would be manifested via job-related uncertainty.

Hypothesis 3: Job-related uncertainty is negatively related to control.

Hypothesis 4: Control is negatively related to psychological strain.

MANAGING UNCERTAINTY DURING ORGANIZATIONAL CHANGE

Management communication is one of the most commonly used and advocated strategies in reducing employee uncertainty during change (Klein, 1996; Lewis, 1999; Lewis & Seibold, 1998; Schweiger & Denisi, 1991). There are two ways in which communication may serve to reduce potential negative outcomes of the change process (Bordia et al, 2001). First, the content or quality of the management communication enables employees to gain change-related information, helping them to feel more prepared and able to cope with change. Second, the participatory nature of the communication process allows employees to participate in decision making, thereby increasing their awareness and understanding of the change events and providing them with a sense of control over change outcomes (Locke & Schweiger, 1979).

Quality of Change Communication (QCC)

Change communication can provide information that helps people understand and deal with the change process (Lewis & Seibold, 1998). Several authors have noted the link between communication and employee uncertainty (Bastien, 1987; DiFonzo & Bordia, 1998; Richardson

& Denton, 1996; Schweiger & DeNisi, 1991; Terry et al., 1996). For example, Schweiger and DeNisi (1991) found that uncertainty was lower in a group that was provided with a systematic program of communication; that is, communication that is timely, credible and trustworthy reduces uncertainty and equips employees with necessary information to deal with organizational change (Bastien, 1987; Richardson & Denton, 1996). Therefore, we predicted that QCC would be negatively related to all types of uncertainty.

Hypothesis 5: QCC is negatively related to strategic uncertainty.

Hypothesis 6: QCC is negatively related to structural uncertainty.

Hypothesis 7: QCC is negatively related to job-related uncertainty.

Communication may also lead to increased feelings of personal control. Research in newcomer socialisation has noted that individuals engage in information and feedback seeking behaviours in order to gain feelings of clarity and personal control during organisational entry (Ashford & Black, 1996). Thus, the receipt of information is critical to the development of control. Applied to a change setting, the information provided about change related issues helps increase an individual's knowledge and understanding of the change and its consequences. With this increased understanding, people are better equipped to deal with future events, which instils a sense of control (Miller, 1981). This implies that the control-inducing effects of communication are largely due to uncertainty reduction. Therefore, we predicted that uncertainty would mediate the effects of communication upon control and that there would be no direct effects of change communication on control.

Participation in Decision-Making (PDM)

Participation in decision making is defined as a process in which influence or decision-making is shared between superiors and their subordinates (Sagie, Elizur, & Koslowsky, 1995). PDM is a communicative activity (Miller, Ellis, Zook & Lyles, 1990) but levels of participation may vary from one context to another (Locke & Schweiger, 1979). For example, participation may be forced or voluntary, formal or informal, direct (individual participation) or indirect (representation on committees) and full authority or minimal consultation (Locke & Schweiger, 1979; Ganster & Fusilier, 1989). These differences suggest that the *effects* of participation may depend upon the *degree* of participation (Ganster & Fusilier, 1989). Furthermore, different types of decisions may be discussed in participation efforts. These include strategic decisions, such as whether the organisation should be changed, and tactical decisions, such as when, where and how to implement the change (Sagie & Kowslowsky, 1996; Sagie, Elizur, & Koslowsky, 1990, 1995).

Reviews of the participation literature have indicated that the effect of PDM on work attitudes is generally positive (Locke & Schweiger, 1979; Schweiger & Leana, 1986; Spector, 1986). It has been shown that when employees are involved in the implementation of new programs, they are more likely to perceive the program as being beneficial (Graham & Verma, 1991; Coyle-Shapiro, 1999). Employee involvement in tactical decisions (compared to strategic decisions) has been found to lead to employee acceptance of or openness toward change (Sagie et al., 1990, 1995; Sagie & Koslowsky, 1996; Wanberg & Banas, 2000) and improved attitudes (Sagie & Koslowsky, 1996). The process by which PDM improves attitudes has been found to be complex, involving numerous mediating variables (e.g., control, change acceptance; Sagie & Koslowsky, 1996).

PDM, like communication, is associated with reduced levels of uncertainty. Macy, Peterson and Norton (1989) found that individuals involved in participation reported higher levels of clarity regarding decision-making criteria. Employee involvement yields positive attitudes because of the reduction in ambiguity and uncertainty (Jackson, 1983; Sagie & Koslowsky, 1996), and the increased levels of knowledge about decisions (Miller & Monge, 1986). Thus, we predicted that PDM would be associated with reduced uncertainty about strategic, structural, and job-related issues.

Hypothesis 8: PDM is negatively related to strategic uncertainty.

Hypothesis 9: PDM is negatively related to structural uncertainty.

Hypothesis 10: PDM is negatively related to job-related uncertainty.

PDM is associated with increased levels of control (e.g., Ganster & Fusilier, 1989; Sagie & Koslowsky, 1996), especially when participation involves discussions of something meaningful and relevant to employees, such as tactical issues. For example, Jackson (1983) and Sagie et al. (1995) found that participation resulted in higher levels of perceived influence. Similarly, Macy et al. (1989) found that direct participation was significantly associated with higher levels of influence over resources, work activities, co-ordination and work time. These results suggest that being actively involved in decision-making is positively associated with control over work issues.

Hypothesis 11: PDM is positively related to control.

PDM is linked with reduced levels of physical and psychological stress (Caplan, Cobb, French, Harrison & Pinneau, 1980; Jackson, 1983; Miller et al., 1990; Spector, 1986). For example, Spector (1986) in a meta-analysis found that participation was correlated negatively with physical symptoms and emotional distress (e.g., anxiety, depression). Similarly, Caplan et al. (1980) found that employee participation in decision-making was significantly negatively correlated with a range of behavioural stress-related outcomes (e.g., number of cigarettes smoked). In a longitudinal field experiment, Jackson (1983) manipulated participation by assigning hospital employees to either a no-intervention or participation group. The intervention was aimed at increasing employees' influence over work-related issues. Jackson found that the intervention led to a significant reduction in emotional strain. These studies suggest that increasing employees' involvement in decision making is effective in reducing job-related strain. Therefore, we made the following prediction.

Hypothesis 12: PDM is negatively related to psychological strain.

THE PROPOSED MODEL

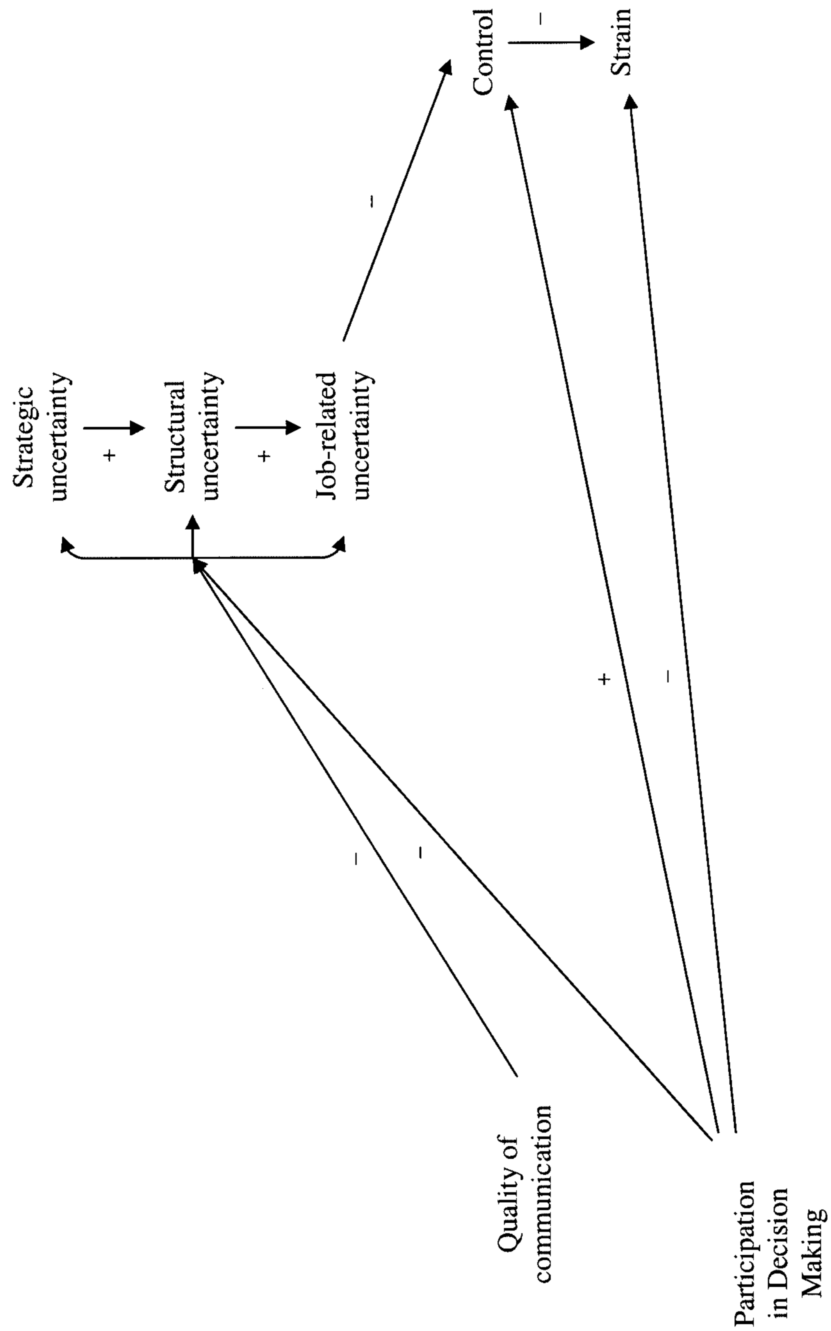
Figure 1 shows the theoretical model representing the 12 hypotheses developed above. We propose that the three types of uncertainties during organizational change are related in a cascade-like fashion, such that, strategic uncertainty leads to structural uncertainty (H1), which in turn, leads to job-related uncertainty (H2). Job-related uncertainty is negatively related to control (H3). Control is negatively related to psychological strain (H4). Thus, control mediates the job-related uncertainty–psychological strain relationship. The model also makes predictions about the effects of uncertainty management strategies. QCC is negatively related to strategic, structural and job related uncertainties (H5, H6, & H7, respectively). PDM has more wide-ranging benefits. It is negatively related to strategic (H8), structural (H9), and job-related (H10) uncertainty, positively related to control (H11), and negatively related to strain (H12). In the following sections, we present the methodology and results of an empirical test of this model.

METHOD

Participants and Procedure

Data were collected from a state government department that had recently been separated (or de-merged) from another government depart-

Figure 1
The Hypothesized Model, Indicating Direction of Relationships



ment involved in the delivery of various services state-wide. This demerger had led to internal restructuring (de-mergers of existing work units, relocation of staff, creation of new work units, changes in job roles and reporting relationships). When the survey was conducted, nearly all the restructuring was complete, and the other parts of the department were now re-established in a new department in a separate building. However, as a result of recent state-level elections, the organization was now dealing with a new government and changes to funding sources, further internal restructuring, and under a new CEO, a redefinition of the core mission and business strategy. The possibility of these changes had been foreshadowed by the new CEO to the staff. Staff communication and consultation was underway. During this period of restructuring and change, the CEO and senior management team were interested in obtaining employee opinions regarding their change-related leadership, sources of staff concern, and the effectiveness of the change communication program and in monitoring a range of employee well-being indicators (such as uncertainty, control, and psychological strain).

Surveys were mailed to all employees of the organization (N=1283). A cover letter from management explained the purpose of the survey and requested their participation. Respondents returned their survey in a pre-paid envelope addressed to the research team. A total of 877 employees returned completed surveys, leading to a response rate of 68.4%. A wide range of age groups (under 20: 2.3%; 20–25: 10.8%; 26–30: 18.8%; 31–35: 15.6%; 36–40: 13.3%; 41–45: 11.7%; 46–50: 13%; over 50: 13.8%; missing: 0.6%) were represented. There were 53% females and 47% males in the sample.

Measures

The survey contained scales for each of the variables in the model.

Quality of Change Communication (QCC). QCC was measured by a 6-item scale developed from recommendations in the change communication literature on characteristics of effective change communication and previous measures of quality of change communication (Miller, Johnson, & Grau, 1994; Miller & Monge, 1985). Respondents were asked to rate the change communication on various dimensions, such as usefulness, timeliness and accuracy (1 = strongly disagree to 6 = strongly agree).

Participation in Decision-Making (PDM). PDM was measured by a 4-item scale. The items were adapted from existing measures of PDM (Hodson, Creighton, Jamison, Rieble, & Welsh, 1994; Miller et al., 1990). The items tapped both participation (e.g., I actively participate in decision-making regarding things that affect me at work) and a sense that the participation mattered (e.g., At work, my ideas and opinions are valued and paid attention to).

Uncertainty. We were interested in measuring the three types of uncertainties: strategic, structural and job-related. We developed 4 item measures for each type. The items were adapted from previous measure of uncertainty during change (Schweiger & Denisi, 1991) and developed from our definitions of each type of uncertainty. Respondents were asked to indicate their uncertainty (1 = very uncertain to 7 = very certain) on the following topics: (1) the ability of the organization to meet the future needs of its customers, the direction in which the organization is heading, the business environment in which the organization will have to exist, the overall objective/mission of the organization (strategic uncertainty); (2) whether work units in the organization will be re-organized in the future, the existing reporting structures (i.e., the chain of command) in the organization, the role/function of different work units within the organization, how your work unit contributes to the overall mission of the organization (structural uncertainty); (3) whether you will have to learn new job skills, the extent to which your job role/tasks will change, the future of your position in the organization, what you need to do to advance within the organization (job-related uncertainty).

Control. Previous research on control during change has operationalized control as an individual difference variable and measured it using locus of control measures (Ashford, 1988; Callan, Terry, & Schweitzer, 1994). Our conceptualization of control, however, involves a situational appraisal by an employee of his or her ability to control the impact of organizational change upon work life. Therefore, we adapted previous measures of control (Bordia et al., 2001; Terry & O'Leary, 1995) and developed a global measure of control over the work environment. The items described control over the employee's future in the organization, nature of changes in the work unit and the direction in which the employee's career is headed (1 = strongly disagree to 7 = strongly agree).

Psychological Strain. Psychological strain was measured by 6 items from Goldberg's (1972) general health questionnaire. Respondents were asked to indicate the extent to which they had recently experienced psychological strain symptoms (e.g., been able to concentrate on what you are doing; 1 = more than usual to 4 = much less than usual).

RESULTS

Table 1 provides the means, standard deviations, inter-correlations and internal consistency alphas for all of the variables. The scale means

Table 1
Means, Standard Deviations (SD), Inter-Correlations, and Internal Consistency Alphas for the Study Variables

	Mean	SD	1	2	3	4	5	6	7
1. Quality of Change Communication	3.83	.97	(.94)[@]						
2. Participation in Decision-Making	4.67	1.53	.45	(.94)					
3. Strategic Uncertainty	3.85	1.19	–.51	–.28	(.81)				
4. Structural Uncertainty	3.35	1.24	–.41	–.31	.58	(.79)			
5. Job-related Uncertainty	4.13	1.41	–.37	–.38	.41	.39	(.68)		
6. Control	3.91	1.26	.40	.58	–.38	–.35	–.40	(.87)	
7. Psychological Strain	2.14	.51	–.22	–.31	.16	.14	.22	–.36	(.84)

Note. N = 877. All correlations are significant at $p < .001$, two-tailed.
[@]Internal consistency alphas are in parentheses along the diagonal.

show that the assessment of the quality of communication, PDM, uncertainty, control, and strain were around the mid-point of the respective scales. Interestingly, the mean level of job-related uncertainty (4.13) was higher than both strategic uncertainty (3.85), paired-sample t (876) = 5.79, $p < .001$, and structural uncertainty (3.35), paired-sample t (876) = 15.66, $p < .001$. Further, mean level of strategic uncertainty was higher than structural uncertainty, paired-sample t (876) = 13.15, $p < .001$. All bivariate correlations were statistically significant and in the expected direction. For example, participation in decision-making was positively related to control, but negatively related to psychological strain. The correlations were moderate in size (the highest being .58 between strategic and structural uncertainty). The internal consistency alphas were all above .79, except for job uncertainty (.68).

Self-reported data from a one-shot survey can contribute to inflated relationships between variables due to common method variance (CMV; Podsakoff & Organ, 1986). We conducted the Harman's one-factor test to see the extent to which CMV was a concern. A factor analysis was conducted on all items measuring the seven variables. A seven-factor solution was obtained (using the eigenvalue greater than one criterion), which explained a total of 68.3% of the variance. The first factor only accounted for 32.9% of the variance. Given that no single factor emerged explaining majority of the variance and seven distinct factors were found, CMV does not seem to be a significant threat to the obtained results.

To test the predicted model in Figure 1, we used structural equation modelling. In this technique, the predicted covariance matrix is compared with the obtained covariance matrix. The match between the matrices is measured by a chi-square test. A significant chi-square indicates that the predicted covariance matrix is significantly different from that obtained. Due to the fact that a significant chi-square is usually obtained with large sample sizes, it is generally recommended that a chi-square value that is less than three times the degrees of freedom (chi-square/df < 3) indicates a good fit of the model. Other indicators of fit include the Tucker-Lewis Index (TLI: Bentler & Bonett, 1980), Comparative Fit Index (CFI; Bentler, 1990); and the Root Mean Square Error of Approximation (RMSEA; Browne & Cudeck, 1993). TLI and CFI range between 0 to 1, with values of .90 and above representing a good fit. An RMSEA value of .05 or less is indicative of a good fit.

The data were analysed using the Analysis of Moment Structures program (AMOS Version 4.0). We used the two-step approach recommended by Anderson and Gerbing (1988). The first step involved the development and test of the measurement model. This allowed us to assess the discriminant validity of the measures used in the survey. The second step involved the test of the structural model predicted in Figure 1.

Test of the Measurement Model

The measurement model had an adequate fit (chi-square = 1699.02, df = 506, $p < .001$; chi-square/df = 3.36; TLI = .93; CFI = .94; RMSEA = .05). However, some of the observed variables had highly correlated error terms. To ensure the unidimensionality of measurement (Anderson & Gerbing, 1988) and to improve the fit of the model, one item was removed from each of the following scales: Strategic uncertainty ("about the ability of the organization to meet the future needs of its customers"), structural uncertainty ("about whether work units in the organization will be re-organized in the future?"), control ("I feel in control of the direction in which my career is headed"), and psychological strain ("felt that you are playing a useful part in things?"). In addition, the measure of job uncertainty was modified by removing two items ("about whether you will have to learn new job skills" and "about the extent to which your job role/tasks will change"). These modifications led to a significant improvement in the fit of the model (chi-square = 764.53; df = 329, $p < .001$, chi-square/df = 2.32; chi-square difference = 934.49, df difference = 177, $p < .001$; TLI = .97; CFI = .97, RMSEA = .04). The standardised path coefficients for each of the items are provided in Table 2. All the indicator variables had high path coefficients from their latent factor. This, in addition to high internal consistency alphas and moderate intercorrelation values among the latent variables, provides confidence in the measurement scales used in this study.

Test of the Structural Model

Step two of the analysis involved testing the structural model (Model 1) hypothesized in Figure 1. The model had a good overall fit to

Table 2
Standardized Path Coefficients from the Measurement Model

Item	Path Coefficients
Quality of Change Communication	
1. Has been useful.	.86
2. Has adequately answered my questions about the changes.	.86
3. Has been positive.	.84
4. Has been communicated appropriately.	.87
5. Has been timely.	.83
6. Has been accurate.	.83
Participation in Decision-Making	
1. I am allowed to provide input on decisions regarding my job role.	.84
2. My supervisor seeks my input on important decisions.	.89
3. At work, my ideas and opinions are valued and paid attention to.	.93
4. I actively participate in decision-making regarding things that affect me at work.	.89
Strategic Uncertainty	
1. About the direction in which the organization is heading?	.82
2. About the business environment in which the organization will have to exist?	.68
3. About the overall objective/mission of the organization?	.81
Structural Uncertainty	
1. About the existing reporting structures (ie the chain of command) in the organization?	.69
2. About the role/function of different work units within the organization?	.74
3. About how your work unit contributes to the overall mission of the organization?	.80
Job-Related Uncertainty	
1. About the future of your position in the organization?	.72
2. About what you need to do to advance within the organization?	.73
Control	
1. I feel I am in control of my future in the organisation.	.71
2. I feel I can influence the nature of change in my work unit.	.78
3. I feel in control of issues related to my job.	.77
4. What I do in this organization is largely under my control.	.75
5. I can influence the extent to which changes at work affect my job.	.80
Strain	
1. Been able to concentrate on what you are doing?	.61
2. Felt capable of making decisions about things?	.60
3. Been able to enjoy your normal day to day activities?	.83
4. Been feeling reasonably happy all things considered?	.88
5. Been able to face up to your problems?	.65

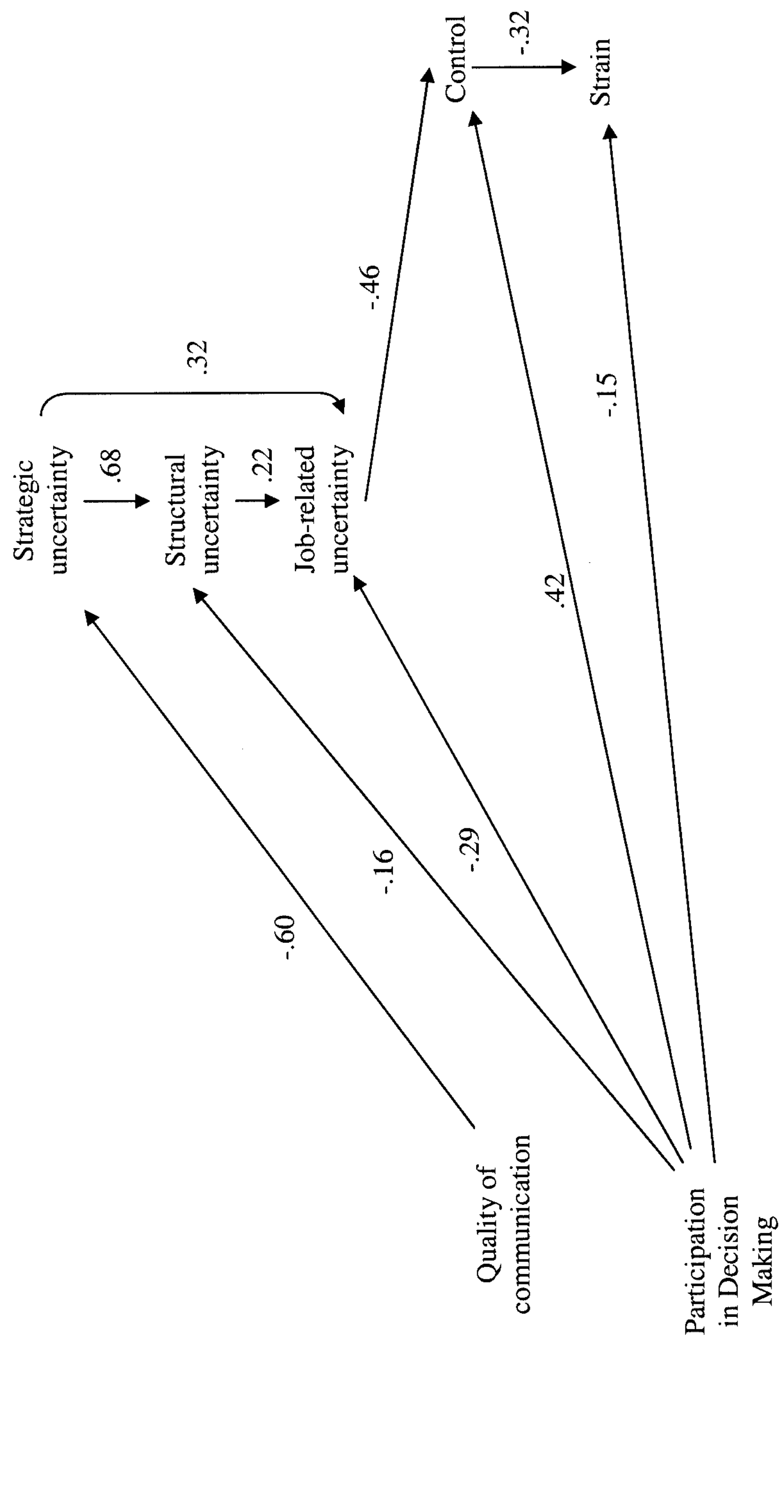
the obtained covariance matrix (chi-square = 785.46; df = 336; $p < .001$; chi-square/df = 2.34; TLI = .97; CFI = .97, RMSEA = .04). Most of the predicted paths were significant and in the expected direction. However, the paths between QCC and structural uncertainty, QCC and job-related uncertainty, and participation and strategic uncertainty were not significant.

As suggested by Kelloway (1995), a mediation-based model should be compared with a model that includes direct paths. We created another model (Model 2) which included the following direct paths: 1) strategic uncertainty to job uncertainty, 2) strategic uncertainty to control, 3) strategic uncertainty to psychological strain, 4) structural uncertainty to control, 5) structural uncertainty to psychological strain, 6) job uncertainty to psychological strain, and 7) QCC to control. When compared to Model 1, this model led to a significant improvement in fit (chi-square = 765.91, df = 330, chi-square/df = 2.32; $p < .001$; chi-square difference = 19.55, $p < .01$; df difference = 6; TLI = .97; CFI = .97; RMSEA = .04). However, except for the strategic uncertainty to job-related uncertainty path, all the added paths were not significant. As a final step, we developed and tested a model (Model 3) by removing all the non-significant paths. This model had a good overall fit (chi-square = 778.26, df = 339, chi-square/df = 2.30, TLI = .97, CFI = .97, RMSEA = .038) and was accepted as the final model. Figure 2 presents this model with the standardized path coefficients. All individual paths were statistically significant and in the hypothesized direction.

DISCUSSION

In this study, we developed and tested a model of types of uncertainties during change, their consequences, and the effects of uncertainty management strategies. Overall, the results supported the predicted model. As predicted by Hypotheses 1 and 2, strategic uncertainty was positively related to structural uncertainty, and structural uncertainty was positively related to job-related uncertainty. Furthermore, the final model indicated that strategic uncertainty was positively related to job uncertainty. Although this latter result was not predicted, taken as a whole the results support previous discussions that different types of uncertainty influence each other (Buono & Bowditch, 1989; Jackson et al., 1987). These inter-relationships highlight the impact of strategic and structural changes upon uncertainties at the individual job level. Indeed, the level of job-related uncertainty was higher than strategic or structural uncertainty. The organization had recently completed a deamalgamation and experienced a change in government, both of which contributed to high levels of organizational instability. This was charac-

Figure 2
The Final Model



Note: All paths are significant at $p \leq .001$.

terized by uncertainty about the business strategy, identity and role of the separate department, the new reporting relationships within the department, and different task and job demands upon employees. Employees were aware that the change in government signaled the need for the department to realign its business strategy and reanalyze the organisation's work processes. In this context of heightened ambiguity, employees were experiencing all three types of uncertainties.

As predicted by Hypothesis 3, job-related uncertainty was negatively related to control. In addition, job-related uncertainty was found to mediate the effects of strategic and structural uncertainty upon control. These results extend previous findings that uncertainty is a precursor of feelings of lack of control, by suggesting that uncertainty about role and task issues is the main contributor to global perceptions of loss of control during change (Callan, 1993; Terry & Jimmieson, 1999). Strategic and structural uncertainty contribute to job-related uncertainty, however, it is the latter type of uncertainty that has the most profound impact on employees' ability to deal with the organizational change. Employee resistance to change is often linked to changes to perceived opportunities for promotion and changes in status due to modifications in one's job role (Callan, 1993; Piderit, 2000).

Control was negatively related to psychological strain (H4) and mediated the effects of uncertainty upon psychological strain. This finding supports a long-standing theoretical assumption, but one that has only recently been empirically demonstrated in the context of organizational change (Bordia, et al., 2001). Previous research has identified uncertainty as an aversive and stressful psychological state (Ashford, 1988; Schweiger & DeNisi, 1991). Our results add to this finding by demonstrating that what makes uncertainty stressful is its negative effect upon control. This finding has theoretical and practical significance. It delineates the psychological mechanism of uncertainty's stressful consequences. Thus, it provides a better understanding of employee reactions to managers wanting to help them deal with organizational change. In addition to efforts at reducing uncertainty, management should include strategies aimed at enhancing employees' feelings of control over their work environment to help them adjust to a dynamic work environment.

In support of Hypothesis 5, we found a negative relationship between QCC and strategic uncertainty. Hypotheses 6 and 7 were not supported, however. QCC was not related to structural and job-related uncertainty. PDM demonstrated a different pattern of relationships; contrary to Hypothesis 8, PDM was not related to strategic uncertainty. Instead, as predicted by Hypotheses 9 and 10, it was negatively related to structural and job-related uncertainty. Further, as predicted, PDM was positively related to control (H11) and negatively related to psychological strain (H12). This pattern of results has important implications for change implementation. The results suggest that management communication is effective in reducing uncertainty about strategic aspects of the change. One-way communication channels may be appropriate for communicating information about upper management decisions that are typically strategic in nature. However, to reduce feelings of uncertainty regarding structural and job-related issues participative strategies, such as team meetings, are required. By being involved in and contributing to decision-making, employees experience less uncertainty about issues affecting them, and feel more in control of change outcomes. This idea is consistent with previous research by Jackson (1983) and Macy et al. (1989), who found that participation was positively associated with control and influence over work issues. The results are consistent with Sagie and colleagues' (1990, 1995, 1996) findings that improved work attitudes and change acceptance can result from participation in tactical issues, as this is how individuals can satisfy their own interests and concerns. It appears that employees require more involvement and participation when the issues are of high relevance to them, and this serves to enhance their feelings of personal control. Similarly, the negative association between control and psychological strain supports previous findings that control has advantageous effects for employee well-being (e.g., Barnett & Brennan, 1995; Greenberger, Strasser, Cummings, & Dunham, 1989). Overall, these results highlight the importance of involving employees in decisions and ensuring that employees have a sense of control over their work and future within the organization.

Senior managers and those responsible for managing change often express frustration and despair over continued uncertainty among employees, in spite of management communication efforts. Our results suggest that management communication is only effective in reducing strategic uncertainty. This could be due to largely one-way communication practices or because the content of communication only addresses strategic issues. However, reduction in structural and job-related uncertainty is vital to improved psychological adjustment to change. Change is often implemented gradually, with more thought given to strategic reasons for change in the early stages (Klein, 1996; Kotter, 1996). Managers may therefore refrain from commenting on structural and job-related changes because they may themselves be uncertain. However, as soon as they hear about impending change, employees wonder about its impact on the structure of the organization and, more importantly, on their job. If formal communication does not address these uncertainties, informal processes, such as the grapevine, take over. Rumors of lay-offs and downgraded work conditions (loss of benefits, increased workload, etc.) become rampant and compound the communication challenges for managers (DiFonzo & Bordia, 2002; DiFonzo et al., 1994). We recommend an open and participative communication process that keeps employees informed of changes as they happen, but more importantly gives them a sense of control over the impact of the change on their job (DiFonzo & Bordia, 1998). In addition to the uncertainty reduction and control inducing effects of PDM, there are other advantages to be derived from participatory change implementation. For example, participation is positively associated with perceptions of fairness, which is vital for acceptance of change and long-term commitment to organizational goals. This may be especially true during times of uncertainty. Van den Bos (2001) has recently demonstrated that fairness of treatment is more important when uncertainty is high. Having a say in matters that affect them and knowing that the change processes will be managed fairly may provide employees the resilience needed to cope with uncertainty.

Limitations and Future Directions

A cross-sectional survey was used to collect data for model testing. As a result, causal inferences cannot be drawn. Future research should separate the measurement of variables over time. In addition, our final measure of uncertainty had only two items measuring job-related uncertainty. While this may have been sufficient to test our theoretical model, future research should develop more items to allow measurement of the different job-related uncertainties experienced by employees during organizational change.

This research was conducted in a public sector organization. Public and private sector organizations differ in their business environment, management practices, and staff attitudes (Bordia & Blau, 1998; Fottler, 1981). However, organizations in both sectors have experienced changes and while the content of uncertainty may differ, the psychological experience of uncertainty and its consequences for control and psychological strain should be generalizable across sectors.

The measurement of the quality of change communication and PDM were based on employee self-reports. To assess the validity of employee perceptions, we asked a senior human resources manager who was closely involved in the change-related communication program to rate the quality of change-related communication (1 = very poor quality of change-related communication to 7 = very high quality of change-related communication) and the extent of PDM (1 = no participative decision making to 7 = great deal of participative decision making) in eight separate work units. The ratings provided by the HR manager were correlated with the average ratings obtained from staff in those work units. The two ratings were positively correlated and of moderate strength (quality of change-related communication $r = .38$ and PDM $r = .40$) indicating moderate levels of agreement between the two sources. The moderate strengths of relationships between management practices and employee perceptions are to be expected as there are several moderators affecting perceptions of management actions. These moderators include (but are not limited to): 1) diversity in manager-subordinate relationships even within the same work unit (as noted by the LMX theory of leadership; Liden & Graen, 1980), 2) trust in the manager (Rousseau & Tijoriwala, 1999), and 3) whether the communication meets individual information needs (DiFonzo & Bordia, 1998; Smeltzer, 1991). Future research should strive to collect data on management actions as well as employee perceptions of communication and PDM practices.

Although we have demonstrated the impact of uncertainty upon psychological adjustment, our model could be extended to include change-related outcomes such as openness to and acceptance of change and staff turnover (Jimmieson, 2002; Sagie & Koslowsky, 1996; Sagie et al., 1990, 1995; Wanberg & Banas, 2000). Moreover, our understanding of how employees *cope* with uncertainty during organizational change is limited. For example, while information seeking may be the most common response to uncertainty (Terry et al., 1996), we know very little about the sources of information or how employees go about gathering information. The literature on information seeking by new entrants to organizations (Morrison, 2002; Sias & Wyers, 2001; Teboul, 1994) could be applied to develop insights into information seeking during organizational change. More research is needed into the role of informal communication processes, such as the grapevine, in reducing uncertainty when formal channels are perceived as inadequate (DiFonzo et al., 1994).

CONCLUSION

The recurring cycles of change in organizations can wear employees out (Callan, 1993). In the words of a hospital employee: “I think there is just *so* much change that you think, you just don’t know. I sort of think has that changed? No, that hasn’t changed. Or this has changed? No this hasn’t changed. Oh yes! That’s changed three times and it’s now back to what it was at the beginning!” Uncertainty arising from a changing and dynamic work environment is a major contributor to feelings of change-related psychological exhaustion. Assuming that change will continue to be a feature of organizational life, effective coping with uncertainty will be an important determinant of psychological well-being. Managers will need to devote considerable time and resources in helping employees deal with uncertainty. It is not surprising, then, that the management of uncertainty is noted as a key leadership challenge in today’s organizations (Bennis, Spreitzer, & Cummings, 2001; Clampitt, DeKoch, & Cashman, 2000; Clampitt, DeKoch, & Williams, 2002; Parry, 1999). A prerequisite to helping individuals cope with uncertainty is a better understanding of the psychological nature of and response to uncertainty.

REFERENCES

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, *103*, 411–423.

Armenakis, A. A., & Bedeian, A. G. (1999). Organizational change: A review of theory and research in the 1990s. *Journal of Management*, *25*(1), 293–315.

Ashford, S. J. (1988). Individual strategies for coping with stress during organizational transitions. *Journal of Applied Behavioural Science*, *24*, 19–36.

Ashford, S., & Black, J. (1996). Proactivity during organizational entry: The role of desire for control. *Journal of Applied Psychology*, *81*, 199–214.

Ashford, S. J., Lee, C., & Bobko, P. (1989). Content, causes, and consequences of job insecurity: A theory-based measure and substantive test. *Academy of Management Journal*, *32*, 803–829.

Barnett, R. C., & Brennan, R. T. (1995). The relationship between job experiences and psychological distress: A structural equation approach. *Journal of Organizational Behavior*, *16*, 259–276.

Bastien, D. T. (1987). Common patterns of behavior and communication in corporate mergers and acquisitions. *Human Resource Management*, *26*(1), 17–33.

Bazerman, M. H. (1982). Impact of personal control on performance: Is added control always beneficial? *Journal of Applied Psychology*, *67*, 472–479.

Bennis, W., Spreitzer, G. M., & Cummings, T. G. (2001). *The future of leadership: Today's top leadership thinkers speak to tomorrow's leaders*. San Francisco: Jossey-Bass.

Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, *107*, 238–246.

Bentler, P. M., & Bonett, D. G. (1980). Significance tests and goodness of fit in the analysis of covariance structures. *Psychological Bulletin*, *88*, 588–606.

Berger, C. R., & Bradac, J. J. (1982). *Language and social knowledge: Uncertainty in interpersonal relations*. London: Edward Arnold.

Berger, C. R., & Calabrese, R. J. (1975). Some explorations in initial interaction and beyond: Toward a developmental theory of interpersonal communication. *Human Communication Research*, *1*, 99–112.

Bordia, P., & Blau, G. (1998). Pay referent comparison and pay level satisfaction in private versus public sector organizations in India. *International Journal of Human Resource Management*, *9*, 155–167.

Bordia, P., Hunt, L., Paulsen, N., Tourish, D., & DiFonzo, N. (in press). Communication and uncertainty during organizational change: It is all about control. *European Journal of Work & Organizational Psychology*.

Browne, M. W., & Cudeck, R. (1993). Alternative ways of assessing model fit. In K. A. Bollen, & J. S. Long (Eds.), *Testing structural equation models* (pp. 445–455). Newbury Park, CA: Sage.

Buono, J., & Bowditch, J. (1989). *The human side of mergers and acquisitions*. San Francisco: Jossey-Bass Publishers.

Callan, V. J. (1993). Individual and organizational strategies for coping with organizational change. *Work & Stress*, *7*, 63–75.

Callan, V. J., Terry, D. J., & Schweitzer, R. (1994). Coping resources, coping strategies and adjustment to organizational change: direct or buffering effects? *Work & Stress*, *8*, 372–383.

Caplan, R. D., Cobb, S., French, J. R. P., Harrison, R. V., & Pinneau, S. R. (1980). *Job demands and worker health*. Ann Arbor: MI: University of Michigan, Institute for Social Research.

Carnall, C. A. (1999). *Managing change in organisations*. London: Pearson Education.
Cascio, W. F. (1995). Whither industrial and organizational psychology in a changing world of work? *American Psychologist*, *50*, 928–939.
Clampitt, P. G., DeKoch, R. J., & Cashman, T. (2000). A strategy for communicating about uncertainty. *Academy of Management Executive*, *14*, 41–57.
Clampitt, P. G., DeKoch, R. J., & Williams, M. L. (2002). Embracing uncertaint:. The hidden dimension of growth. *Ivey Business Journal*, *66*(3), 57–67.
Coyle-Shapiro, J. A. M. (1999). Employee participation and assessment of an organizational change intervention. A three-wave study of total quality management. *The Journal of Applied Behavioral Science*, *35*(4), 439–456.
Desveaux, J. A. (1994). Anticipating uncertainty: The strategy-structure problem in public bureaucracy. *Governance: An International Journal of Policy and Administration*, *7*(1), 31–58.
DiFonzo, N., & Bordia, P. (1998). A tale of two corporations: Managing uncertainty during organizational change. *Human Resource Management*, *37*(3&4), 295–303.
DiFonzo, N., & Bordia, P. (2002). Corporate rumor activity, belief and accuracy. *Public Relations Review*, *28*, 1–19.
DiFonzo, N., Bordia, P., & Rosnow, R. L. (1994). Reining in rumours. *Organizational Dynamics*, *23*, 47–62.
Festinger, L. (1954). A theory of social comparison processes. *Human Relations*, *7*, 117–140.
Fottler, M. (1981). Is management really generic? *Academy of Management Review*, *6*, 1–12.
Ganster, D., & Fusilier, M. (1989). Control in the workplace. In C. Cooper, & I. Robertson, (Eds.). *International Review of Industrial and Organizational Psychology*, *4*, 235–280.
Goldberg, D. P. (1972). *The detection of psychiatric illnesses by questionnaire*. Oxford: Oxford University Press.
Graham, J. W., & Verma, A. (1991). Predictors and moderators of employee responses to employee participation programs. *Human Relations*, *44*, 551–568.
Greenberger, D., & Strasser, S. (1986). Development and application of a model of personal control in organizations. *Academy of Management Review*, *11*, 164–177.
Greenberger, D., Strasser, S., Cummings, L., Dunham, R. (1989). The impact of personal control on performance and satisfaction. *Organizational Behavior and Human Decision Processes*, *43*, 29–51.
Greenhalgh, L., & Sutton, R. (1991). Organizational effectiveness and job insecurity. In J. Hartley, D. Jacobsen, B. Klandermans, & T. van Vuuren (Eds.), *Job insecurity: Coping with jobs at risk* (pp. 151–171). London: Sage.
Hodson, R., Creighton, S., Jamison, C. S., Rieble, S., & Welsh, S. (1994). Loyalty to whom? Workplace participation and the development of consent. *Human Relations*, *47*, 895–909.
Hogg, M. A., & Mullin, B. A. (1999). Joining groups to reduce uncertainty: Subjective uncertainty reduction and group identification. In D. Abrams, & M. A. Hogg (Eds.), *Social identity and social cognition* (pp. 249–279). Oxford, UK: Blackwell.
Hui, C., & Lee, C. (2000). Moderating effects of organization-based self-esteem on organizational uncertainty: Employee response relationships. *Journal of Management*, *26*(2), 215–232.
Ito, J. K., & Brotheridge, C. M. (2001). An examination of the roles of career uncertainty, flexibility, and control in predicting emotional exhaustion. *Journal of Vocational Behavior*, *59*, 406–424.
Jackson, S. E. (1983). Participation in decision making as a strategy for reducing job-related strain. *Journal of Applied Psychology*, *68*(1), 3–19.
Jackson, S., Schuler, R., & Vredenburgh, D. (1987). Managing stress in turbulent times. In A. Riley, & S. Zaccaro, (Eds.). *Occupational stress and organizational effectiveness* (pp. 141–166). New York: Praeger.
Jimmieson, N. (2002). *The impact of coping resources on employee adjustment to organizational change: A test of the mediating role of change readiness*. Paper presented at the XXV International Congress of Applied Psychology, Singapore.
Jimmieson, N., & Terry, D. (1999). The moderating role of task characteristics in determining responses to a stressful work simulation. *Journal of Organizational Behavior*, *20*(5), 709–736.

Johnson, J. R., Bernhagen, M. J., Miller, V., & Allen, M. (1996). The role of communication in managing reductions in work force. *Journal of Applied Communication Research*, *24*, 139–164.

Kelloway, E. K. (1995). Structural equation modelling in perspective. *Journal of Organizational Behavior*, *16*, 215–224.

Klein, S. M. (1996). A management communication strategy for change. *Journal of Organizational Change Management*, *9*(2), 32–46.

Kotter, J. (1996). *Leading change*. Boston: Harvard Business School Press.

Kramer, M. W. (1999). Motivation to reduce uncertainty: A reconceptualization of uncertainty reduction theory. *Management Communication Quarterly*, *13*(2), 305–316.

Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. New York: Springer Publishing Company.

Lewis, L. K. (1999). Disseminating information and soliciting input during planned organizational change. *Management Communication Quarterly*, *13*(1), 43–75.

Lewis, L. K., & Seibold, D. R. (1998). Reconceptualizing organizational change implementation as a communication problem: A review of literature and research agenda. In M. E. Roloff (Ed.), *Communication yearbook 21* (pp. 93–151). Beverly Hills, CA: Sage.

Liden, R. C., & Graen, G. (1980). Generalizability of the vertical dyad linkage model of leadership. *Academy of Management Journal*, *23*(3), 451–465.

Locke, E. A., & Schweiger, D. M. (1979). Participation in decision making: One more look. *Research in Organizational Behavior*, *1*, 265–339.

Macy, B. A., Peterson, M. F., & Norton, L. W. (1989). A test of participation theory in a work re-design field setting: Degree of participation and comparison site contrasts. *Human Relations*, *42*, 1095–1165.

Martinko, M. J., & Gardner, W. L. (1982). Learned Helplessness: An alternative explanation for performance deficits. *Academy of Management Review*, *7*, 195–204.

Maurier, W. L., & Northcott, H. C. (2000). Job uncertainty and health status for nurses during restructuring of health care in Alberta. *Western Journal of Nursing Research*, *22*(5), 623–641.

Miller, K., Ellis, B., Zook, E., & Lyles, J. (1990). An integrated model of communication, stress, and burnout in the workplace. *Communication Research*, *17*, 300–326.

Miller, K., & Monge, P. (1985). Social information and employee anxiety about organizational change. *Human Communication Research*, *11*, 365–386.

Miller, K. I., & Monge, P. R. (1986). Participation, satisfaction, and productivity: A meta-analytic review. *Academy of Management Journal*, *29*(4), 727–753.

Miller, S. M. (1981). Predictability and human stress: Toward a clarification of evidence and theory. *Advances in Experimental Social Psychology*, *14*, 203–256.

Miller, V. D., Johnson, J. R., & Grau, J. (1994). Antecedents to willingness to participate in a planned organisational change. *Journal of Applied Communication Research*, *22*, 59–80.

Milliken, F. J. (1987). Three types of perceived uncertainty about the environment: State, effect, and response uncertainty. *Academy of Management Review*, *12*(1), 133–143.

Milliken, F. J. (1990). Perceiving and interpreting environmental change: An examination of college administrators' interpretation of changing demographics. *Academy of Management Journal*, *33*, 42–63.

Morrison, E. W. (2002). Information seeking within organizations. *Human Communication Research*, *28*, 229–242.

Nelson, A., Cooper, C. L., & Jackson, P. R. (1995). Uncertainty amidst change: The impact of privatization on employee job satisfaction and well-being. *Journal of Occupational and Organizational Psychology*, *68*, 57–71.

Orpen, C. (1994). Interactive effects of work motivation and personal control on employee job performance and satisfaction. *Journal of Social Psychology*, *134*(6), 855–856.

Parry, K. W. (1999). Enhancing adaptability: Leadership strategies to accommodate change in local government settings. *Journal of Organizational Change Management*, *12*(2), 134–156.

Piderit, S. K. (2000). Rethinking resistance and recognizing ambivalence: A multidimensional view of attitudes toward an organizational change. *Academy of Management Review*, *25*, 783–794.

Podsakoff, P. M., & Organ, D. W. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management*, *12*(4), 531–544.

Pollard, T. M. (2001). Changes in mental well-being, blood pressure and total cholesterol levels during workplace reorganization: The impact of uncertainty. *Work & Stress*, *15*(1), 14–28.

Putnam, L. L., & Sorenson, R. L. (1982). Equivocal messages in organizations. *Human Communication Research*, *8*(2), 114–132.

Rafferty, A. (2002). *Relating different types of organizational change to measures of employee well-being*. Paper presented at the XXV International Congress of Applied Psychology, Singapore.

Richardson, P., & Denton, D. K. (1996). Communicating change. *Human Resource Management*, *35*(2), 203–216.

Rosnow, R. L. (1991). Inside rumor: A personal journey. *American Psychologist*, *46*, 484–496.

Rousseau, D. M., & Tijoriwala, S. A. (1999). What's a good reason to change? Motivated reasoning and social accounts in promoting organizational change. *Journal of Applied Psychology*, *84*, 514–528.

Sagie, A., Elizur, D., & Koslowsky, M. (1990). Effect of participation in strategie and tactical decisions on acceptance of planned change. *The Journal of Social Psychology*, *130*(4), 459–465.

Sagie, A., Elizur, D., & Koslowsky, M. (1995). Decision type, participative decision making (PDM), and organizational behavior: An experimental simulation. *Human Performance*, *8*(2), 81–94.

Sagie, A., & Koslowsky, M. (1996). Decision type, organisational control, and acceptance of change: An integrative approach to participative decision making. *Applied Psychology: An International Review*, *45*(1), 85–92.

Schweiger, D. M., & DeNisi, A. S. (1991). Communication with employees following a merger: A longitudinal field experiment. *Academy of Management Journal*, *34*(1), 110–135.

Schweiger, D. M., & Leana, C. R. (1986). Participation in decision making. In E. A. Locke (Ed.), *Generalizing from laboratory to field settings* (pp. 147–166). Lexington, MA: Lexington Books.

Sias, P. M., & Wyers, T. D. (2001). Employee uncertainty and information-seeking in newly formed expansion organizations. *Management Communication Quarterly*, *14*(4), 549–573.

Smeltzer, L. R. (1991). An analysis of strategies for announcing organization-wide change. *Group & Organization Studies*, *16*(1), 5–24.

Spector, P. E. (1986). Perceived control by employees: A meta-analysis of studies concerning autonomy and participation at work. *Human Relations*, *39*(11), 1005–1016.

Spector, P. E. (1987). Interactive effects of perceived control and job stressors on affective reactions and health outcomes for clerical workers. *Work & Stress*, *1*, 155–162.

Spector, P. E. (2002). Employee control and occupational stress. *Current Directions in Psychological Science*, *11*(4), 133–136.

Teboul, J. C. B. (1994). Facing and coping with uncertainty during organizational encounter. *Management Communication Quarterly*, *8*(2), 190–224.

Terry, D. J., Callan, V. J., & Sartori, G. (1996). Employee adjustment to an organizational merger: Stress, coping and intergroup differences. *Stress Medicine*, *12*, 105–122.

Terry, D. J., & Jimmieson, N. (1999). Work control and employee well-being: A decade review. *International Review of Industrial and Organizational Psychology*, *14*, 95–148.

Terry, D. J., & Jimmieson, N. (2003). A stress and coping approach to organisational change: Evidence from three field studies. *Australian Psychologist*, *38*, 92–101.

Terry, D. J., & O'Leary, J. E. (1995). The theory of planned behavior: The effects of perceived behavioural control and self-efficacy. *British Journal of Social Psychology*, *34*, 199–220.

van den Bos, K. (2001). Uncertainty management: The influence of uncertainty salience on reactions to perceived procedural fairness. *Journal of Personality and Social Psychology*, *80*(6), 931–941.

Wanberg, C. R., & Banas, J. T. (2000). Predictors and outcomes of openness to changes in a reorganizing workplace. *Journal of Applied Psychology*, *85*(1), 132–142.

Watson, B., Jones, E., Hobman, E., Bordia, P., Gallois, C., & Callan, V. J. (2002). Employee perceptions of organizational change. Manuscript under review.